

Department of Electrical Engineering

विद्युत अभियांत्रिकी विभाग



Electronics Lab

इलेक्ट्रॉनिक्स लैब

Experiment Number 3

प्रयोग संख्या 3

Verification of three terminal voltage regulator IC 7805, 7812, 7905,
7912 and LM317.

तीन टर्मिनल वोल्टेज रेगुलेटर IC 7805, 7812, 7905, 7912
और LM317 का सत्यापन।

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

ELECTRONICS LAB

EXPERIMENT NO.3

Aim :- To study the three terminal voltage regulator IC 7805, 7812, 7912 and LM317.

Instruments required:

1. H.W. Rectifier Unit / Power Supply
2. Multimeter
3. IC 7805, 7812, 7912 & LM317
4. Capacitors (0.33 μ f, 0.01 μ f, 2.2 μ f, 1 μ f)
5. Resistances (240 Ω) and 3K Ω)

Theory:

A voltage regulator is a circuit that supplies a constant regardless of changes in load current. The LM 78XX series of 3-terminal regulators is available with fixed output voltages of +5V, +12V, and 15V. Similarly, 79XX is available for 5V, 12V, and -15V output voltage.

The LM317 series are available with an output voltage of 1.2 to 37V. The LM317 develops a nominal 1.25 V. referred to as the reference voltage V_{ref} between the output and adjustment terminal. Its circuit contains two resistors namely the current set and output set resistor. The maximum value of adjustment pin current I_{adj} is 100 μ A. The output voltage is given by:

$$V_o = V_{ref} (1 + R_L / R_1)$$

Where,

$$V_{ref} = 1.25$$

$$R_1 = \text{Current set resistor.}$$

$$R_L = \text{Output set resistor.}$$

Circuit Diagram:

1. Typical application of IC 7805/7812

Fixed output Regulator



Fig 1.1

Adjustable output Regulator

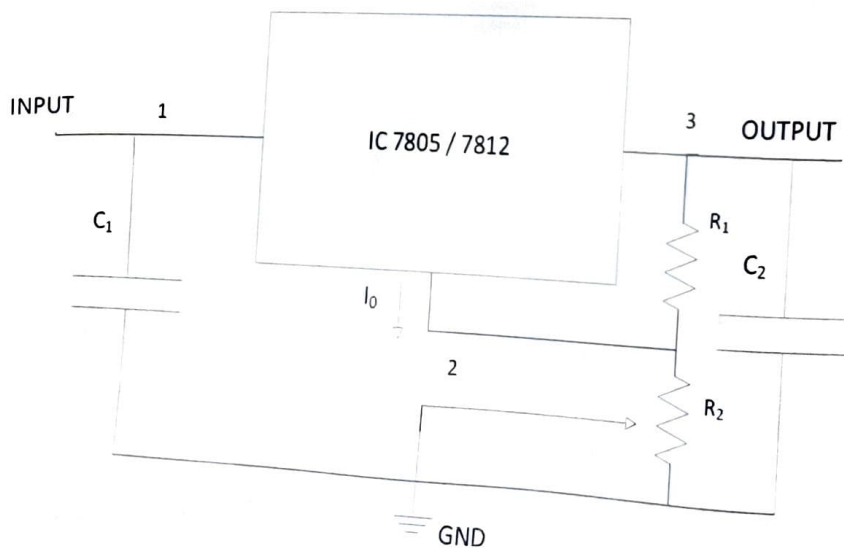


Fig 1.2

2. Typical Application of IC 7912

Fixed Output Regulator

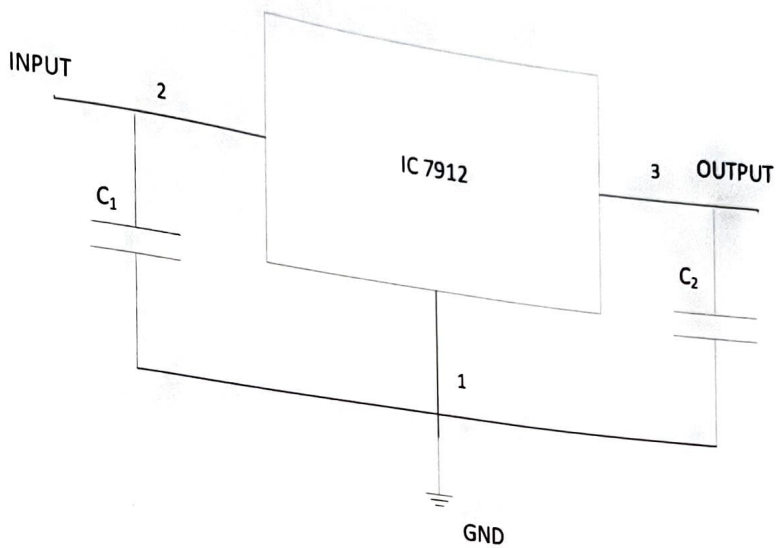


Fig 2.1

Adjustable Output Regulator

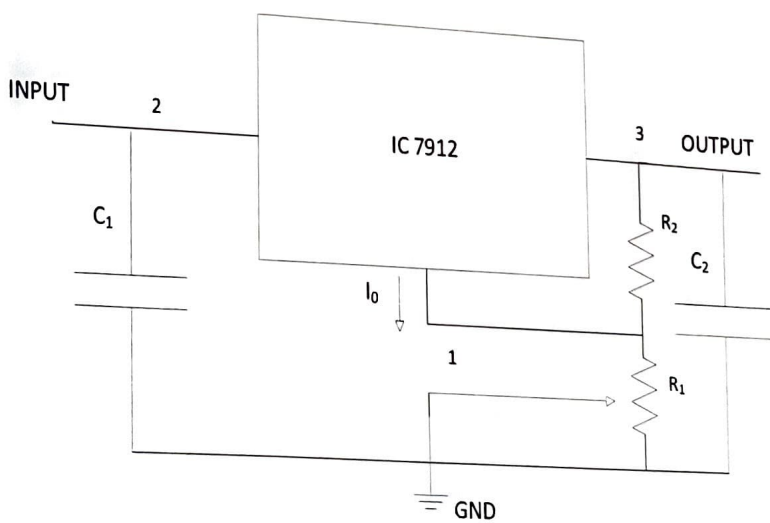


Fig 2.2

3. Typical Application of IC LM317

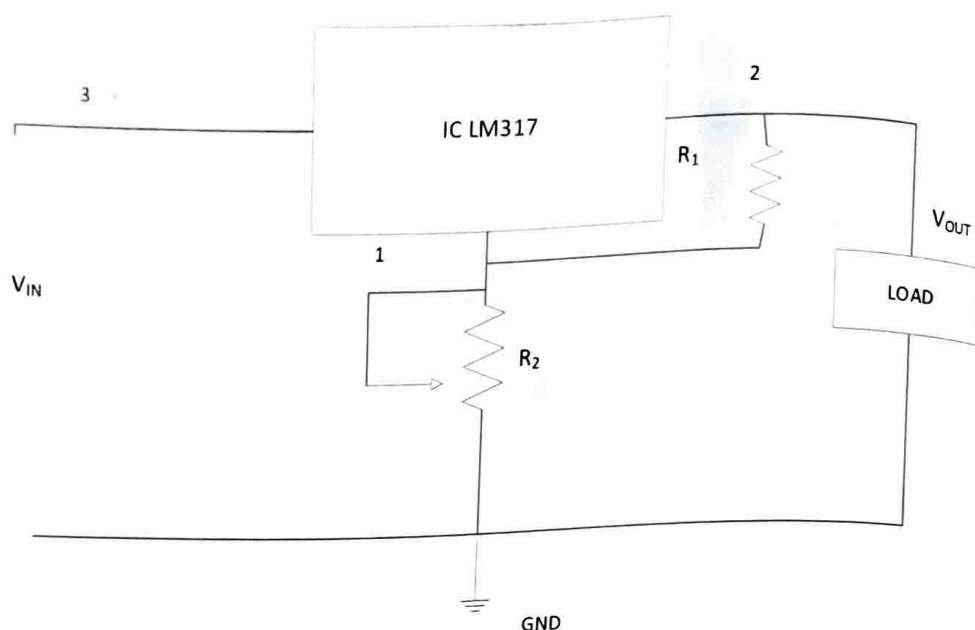


Fig. 3

Procedure:-

Fixed Output Regulator

1. Connect the circuit as per Fig-1.1 & Fig-2.1.2.
2. Give the input of 7V (IC 7805)/14.5V (IC 7812) to Pin No.-1.
3. Give the input of -14.5V (IC 7912) to Pin No.-3.
4. Verify the output voltage for 7805 & 7812 at Pin No.-3 & for 7912 at Pin No.-2.
5. Repeat the above procedure for input-8V, 9V, 10V for 7805, 16V, 17V, 18V, 19V for 7812 & -16V, -17V, -18V, -19V for 7912.

Adjustable Output Regulator:

1. Connect the circuit as per Fig-1.2, Fig-2.2 & Fig-3.
2. Give the input of 10V (IC 7805)/15V (IC 7812) to Pin No. - 1 & 20V (IC317) to Pin No.-2.
3. Give the input of -18V at Pin No.-3 for 7912.

4. Vary the resistance R_a (For 7805, 7812 & 317).
5. Vary the resistance R , (For 7912).
6. Note down and verify the output voltage at Pin No.-3 (For 78XX & 317).
7. Note down and verify the output voltage at Pin No.-2 (For 7912)

Observation table:

S.No.	IC No.	Fixed/Adjustable Output	Input Voltage	Output Voltage	Expected O/P Voltage

Result & Conclusion:

Questions:

1. What is a voltage regulator, and why is it necessary in electronic circuits?
2. Explain the working principle of a three-terminal voltage regulator IC.
3. What are the advantages and disadvantages of using a three-terminal voltage regulator IC like the 7805, 7812, and LM317?
4. What is the output voltage of the 7805 voltage regulator, and for which applications is it typically used?
5. What is the input voltage range required for the 7805 IC to operate correctly?
6. What is unique about the LM317 voltage regulator in comparison to the 7805, 7812, and 7912?
7. What are the typical applications of each of these ICs (7805, 7812, 7912, LM317) in real-world electronics?